

VU Medical Zone

BIO 503 Final Term Subjective

Q1. State fluid mosaic model?

2 Marks

The fluid mosaic model describes the structure of the plasma membrane as a mosaic of components—including phospholipids, cholesterol, proteins, and carbohydrates—that gives the membrane a fluid character.

Q2. Saxitoxin?

2 Marks

Saxitoxin is produced by a dinoflagellate also known as "red tide". It blocks voltage-dependent sodium channels.

Q3. Radiation Protection?

2 Marks

Radiation protection, sometimes known as radiological protection, is defined by the International Atomic Energy Agency (IAEA) as "The protection of people from harmful effects of exposure to ionizing radiation, and the means for achieving this".

Q4. Who discovered DNA? And name of scientist discovered the structure of DNA? 2 Marks

In 1953 Watson and Crick determined the structure of DNA.

Q5. Chromatography?

3 Marks

- Analytical methods used to separate molecules.
- It involves a mobile and a stationary phase.
- Mobile phase is what the material to be separated is dissolved in.
- Stationary phase is a porous solid matrix which the mobile phase surrounds.

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- Separation occurs because of the differing chemistries each molecule has with both the mobile and stationary phase.

Q6. What are hormones? 3 Marks

Hormones are long-distance chemical signals released by endocrine cells (like the cells of your pituitary gland). They control specific physiological processes, such as growth, development, metabolism, and reproduction. While some hormones are steroid-based, others are proteins. These protein-based hormones are commonly called peptide hormones. For example, insulin is an important peptide hormone that helps regulate blood glucose levels.

Q7. Ionizing radiation? 3 Marks

Ionizing radiation – e.g., radiation emissions from uranium ore, and high frequency waves in the electromagnetic spectrum such as x-rays.

Q8. Name 5 methods to reduce friction? 5 Marks

Friction can be reduced by making the surface smooth by polishing.

The friction is due to the roughness of surface. If we make the surfaces smooth by polishing, then friction will be reduced. For example, a slide in the park is polished to make its surface smooth and reduce friction. Due to reduced friction of a polished, smooth slide, children can slide down easily.

Friction can be reduced by applying lubricants (like oil or grease) to the rubbing surface.

When oil or grease is applied between the moving parts of a machine, a thin layer of oil or grease is formed between the two rubbing surfaces. This layer of oil separates the two rubbing surfaces a little bit due to which their interlocking is reduced to a large extent. This friction is reduced and movement becomes smooth. When oil or grease is applied to the moving parts of a machine, then their surfaces do not rub directly against each other, they rub through a layer of oil and grease. The substance which reduces friction are called lubricants.

Friction can be reduced by using wheels to move objects.

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It is quite difficult to move a heavy suitcase by dragging it on the ground because the sliding friction between the heavy suitcase and the ground is very large. If this heavy suitcase is fitted with small wheels, then it can be pulled very easily. Because when we attach wheels then sliding friction is converted into rolling friction. And rolling friction between the wheels of suitcase and ground is much less.

Friction can be reduced by using ball bearing between the moving parts of machine.

Ball bearing is a device which consists of a ring of small metal balls. The small metal balls of a ball bearing can roll freely. They are designed to make the moving parts of a machine to roll over each other rather than slide. The ball bearing is introduced between the two surfaces which have to rotate over each other. The axle is fixed on the inner side of a ball bearing and wheel is fixed to the outer side of the ball bearing. They reduce friction by making the two surfaces (axle and wheel) to roll over each other.

Q9. Explain Na^+/K^+ ATPase?

5 Marks

The Na^+/K^+ ATPase

The cytosol of animal cells contains a concentration of potassium ions (K^+) as much as 20 times higher than that in the extracellular fluid. Conversely, the extracellular fluid contains a concentration of sodium ions (Na^+) as much as 10 times greater than that within the cell. These concentration gradients are established by the active transport of both ions. And, in fact, the same transporter, called the Na^+/K^+ ATPase, does both jobs. It uses the energy from the hydrolysis of ATP to actively transport 3 Na^+ ions out of the cell. For each 2 K^+ ions pumped into the cell. This accomplishes several vital functions:

It helps establish a net charge across the plasma membrane with the interior of the cell being negatively charged with respect to the exterior. This resting potential prepares nerve and muscle cells for the propagation of action potentials leading to nerve impulses and muscle contraction. The accumulation of sodium ions outside of the cell draws water out of the cell and thus enables it to maintain osmotic balance (otherwise it would swell and burst from the inward diffusion of water). The gradient of sodium ions is harnessed to provide the energy to run several types of indirect pumps. The crucial roles of the Na^+/K^+ ATPase are reflected in the fact that almost one third of all the energy generated by the mitochondria in animal cells is used just to run this pump.

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Q10. Explain process of radioactive decay?

10 Marks

Radioactive decay

Radioactive decay (also known as nuclear decay or radioactivity) is the process by which an unstable atomic nucleus loses energy (in terms of mass in its rest frame) by emitting radiation, such as an alpha particle, beta particle with neutrino or only a neutrino in the case of electron capture, gamma ray, or electron in the case of internal conversion. A material containing such unstable nuclei is considered radioactive. Certain highly excited short-lived nuclear states can decay through neutron emission, or more rarely, proton emission.

Radioactivity was discovered in **1896** by the French scientist **Henri Becquerel**, while working with phosphorescent materials.

- These materials glow in the dark after exposure to light, and he suspected that the glow produced in cathode ray tubes by X-rays might be associated with phosphorescence.
- He wrapped a photographic plate in black paper and placed various phosphorescent salts on it. All results were negative until he used uranium salts.
- The uranium salts caused a blackening of the plate in spite of the plate being wrapped in black paper. These radiations were given the name "Becquerel Rays".
- Radioactive decay occurs in unstable atomic nuclei – that is, ones that don't have enough binding energy to hold the nucleus together due to an excess of either protons or neutrons.
- It comes in three main types – named alpha, beta and gamma for the first three letters of the Greek alphabet.

Q11. Define biology.

2 Marks

Biology deals with living matter, collections of atoms and molecules that manage to harness energy to perform such extraordinary tasks Such as - locomotion, - reproduction, - computation ("thinking").

Q12. Why sliding friction less than static friction.

2 Marks

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The friction between the two surfaces is because of the interlocking of the irregularities of the two surfaces.

During sliding, contact points do not get enough time to get interlocked properly. Therefore, less friction. But, when the object is at rest, the surfaces interlock well, therefore friction is more. This explains why sliding friction is less than static friction.

Q13. What is sliding Friction? 2 Marks

Sliding friction occurs between objects as they slide against each other. It acts in the direction opposite to the direction of motion. It prevents the object from moving too fast. It is also called as kinetic friction.

Q14. What is electrolocation? 3 Marks

Electrolocation:

Electroreceptive animals use this sense to locate objects around them. This is important in ecological niches where the animal cannot depend on vision: for example, in caves, in murky water and at night. Many fish use electric fields to detect buried prey.

Some shark embryos and pups "freeze" when they detect the characteristic electric signal of their predators.

Q15. Structure of amino acid. 3 Marks

Structure of Amino Acids Amino acids are organic compounds containing amine ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) functional groups, along with a side chain (R group) specific to each amino acid.

Two amino acids can form a complex through peptide bonds

Amino acids can be covalently joined together in a condensation reaction to form a dipeptide and water. The covalent bond between the amino acids is called a peptide bond and, for this reason, long chains of covalently bonded amino acids are called polypeptides. Polypeptide chains can be broken down via hydrolysis reactions, which requires water to reverse the process.

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Q16. Ion exchange in chromatography. 5 Marks

Ion Exchange Chromatography Ion exchange chromatography involves the separation of ionizable molecules based on their total charge. This technique enables the separation of similar types of molecules that would be difficult to separate by other techniques because the charge carried by the molecule of interest can be readily manipulated by changing buffer pH.

Ion exchange chromatography is commonly used to separate charged biological molecules such as proteins, peptides, amino acids, or nucleotides. The amino acids that make up proteins are zwitterion compounds that contain both positively and negatively charged chemical groups.

Depending on the pH of their environment, proteins may carry a net positive charge, a net negative charge, or no charge. The pH at which a molecule has no net charge is called its isoelectric point, or pI.

The pI value can be calculated based on the primary sequence of the molecule. The choice of buffer pH then determines the net charge of the protein of interest.

Anion exchange: In a buffer with a pH greater than the pI of the protein of interest, the protein will carry a net negative charge; therefore, a positively charged anion exchange resin is chosen to capture this protein.

Cation exchange: In a buffer with a pH lower than the pI of the protein of interest, the protein will carry a positive net charge; thus, a negatively-charged cation exchange resin is chosen.

Q17. What is active and passive electrolocation? 5 Marks

Active electrolocation:

In active electrolocation, the animal senses its surrounding environment by generating electric fields and detecting distortions in these fields using electroreceptor organs. This electric field is generated by means of a specialized electric organ consisting of modified muscle or nerves. This field may be modulated so that its frequency and wave form are unique to the species and sometimes, the individual.

Animals that use active electroreception include the weakly electric fish, which either generate small electrical pulses ("pulse-type") or produce a quasisinusoidal discharge from the electric organ (termed "wave-type"). These fish create a potential which is usually smaller than one volt. Weakly electric fish can discriminate between objects with different resistance and capacitance

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values, which may help in identifying the object. Active electroreception typically has a range of about one body length, though objects with an electrical impedance similar to that of the surrounding water are nearly undetectable.

Passive electrolocation:

In passive electrolocation, the animal senses the weak bioelectric fields generated by other animals and uses it to locate them. These electric fields are generated by all animals due to the activity of their nerves and muscles. A second source of electric fields in fish is the ion pumps associated with osmoregulation at the gill membrane. This field is modulated by the opening and closing of the mouth and gill slits. Many fish that prey on electrogenic fish use the discharges of their prey to detect them. This has driven the prey to evolve more complex or higher frequency signals that are harder to detect.

Passive electroreception is carried out solely by ampullary electroreceptors in fish. It is tuned to low frequency signals (below one up to tens of Hertz). Fish use passive electroreception to supplement or replace their other senses when detecting prey and predators. In sharks, sensing an electric dipole alone is sufficient to cause them to try to eat it.

Q18. Write down the function of cell membrane.

5 Marks

Functions of the cell membrane:

To maintain the physical integrity of the cell – that is to mechanically enclose the contents of the cell, and also. To control the movement of particles e.g., ions or molecules, into and out of the cell. The cell membrane maintains the physical integrity of the cell. It's most obvious in the cases of animal cells (because they don't have cell walls) that the cell membrane holds the cell together by enclosing the cytoplasm and organelles within it. The cell membrane forms a barrier between the inside of the cell and the environment outside the cell – enclosing cytoplasm and any organelles within the cell, and enabling different chemical environments to exist on each side of the cell membrane. The cell membrane physically separates the intracellular components (e.g., organelles in eukaryotic cells) from the extracellular environment. The cell membrane protects the cell from some harmful chemicals in its external environment. It also protects the cell from loss of useful biological macromolecules held within the cell by its plasma membrane.

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